

INTERNSHIP PROJECT REPORT

Task 3

House Price Prediction using Model Validation and Hyperparameter Tuning

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Objective, Dataset and Exploratory Analysis Objective:

To improve the performance of a house price prediction model by applying model validation techniques, detecting overfitting, performing cross-validation, and tuning hyperparameters using GridSearchCV to select the best regression model.

Dataset and Workflow:

Dataset: California Housing Dataset (20,640 records, 8 input features and 1 target variable).

Workflow: Data loading → inspection → EDA → train/test split → feature scaling → regression models → cross-validation → overfitting check → GridSearchCV → final model selection.

Exploratory Data Analysis:

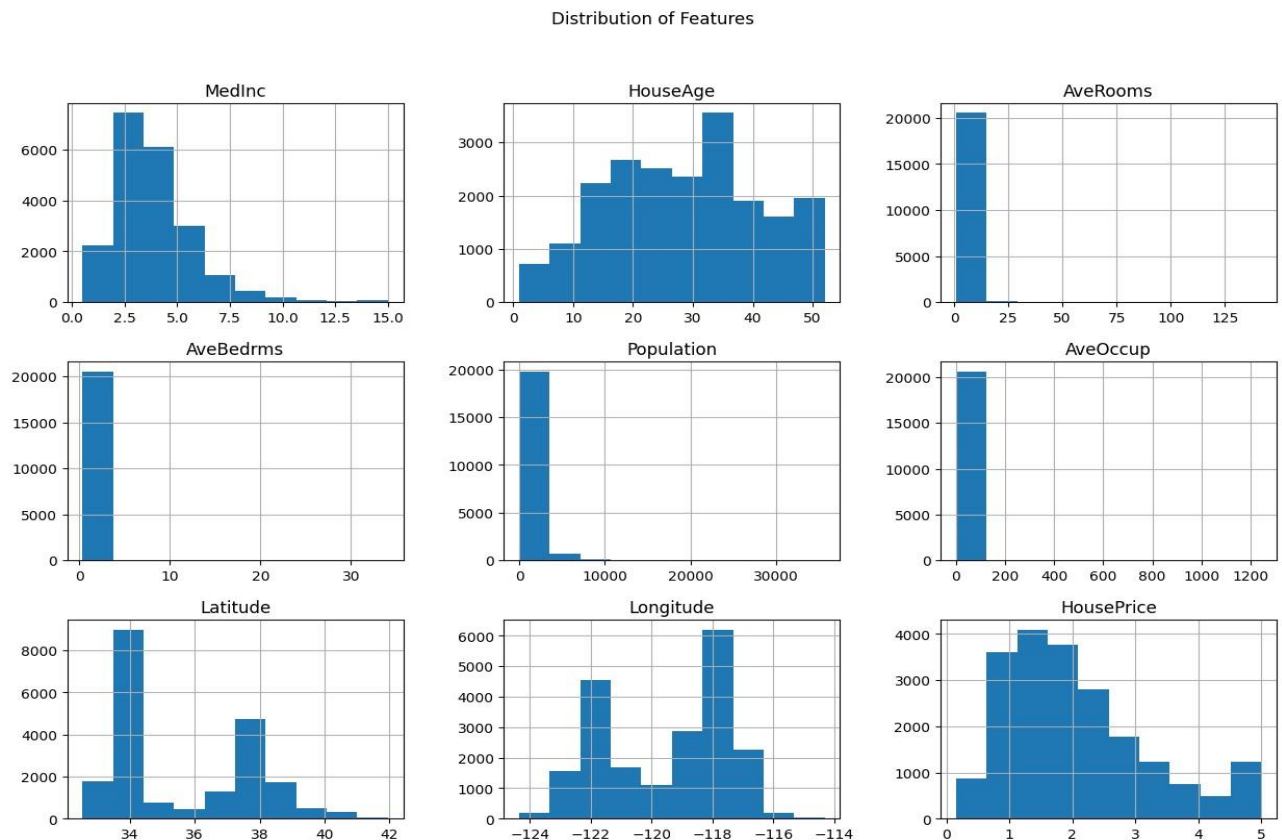
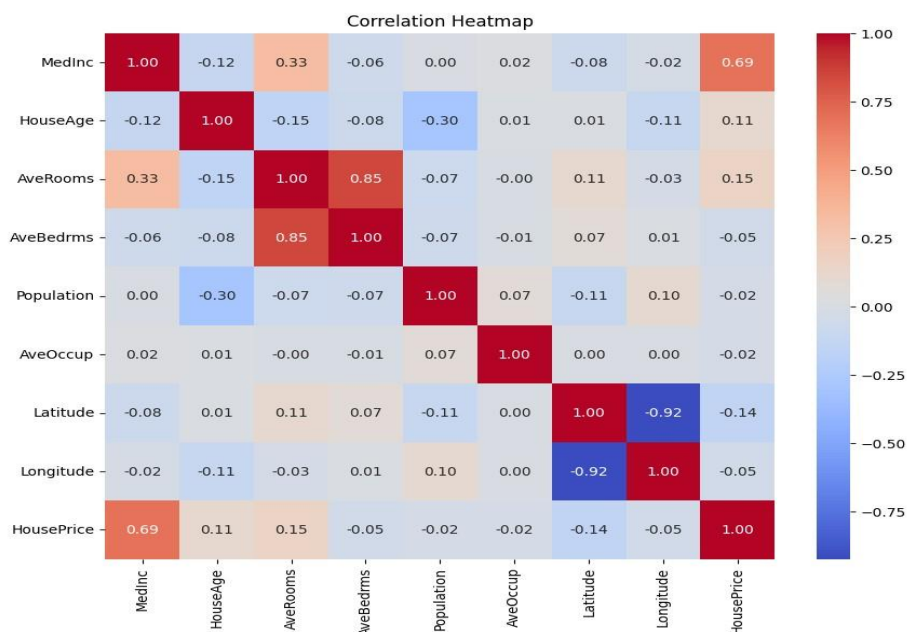


Figure 1. Distribution of the dataset features.

The notebook checks the dataset structure, descriptive statistics and missing values before modelling. The dataset contains 20,640 observations and the required features are available for training.

Model Development and Initial Results:

Three regression approaches were implemented: Linear Regression, Ridge Regression and Decision Tree Regression. The data was split into 80% training and 20% testing data.



Initial model comparison:

Model	RMSE	R ²
Linear Regression	0.745581	0.575788
Ridge Regression	0.745522	0.575855
Decision Tree	0.703729	0.622076

The initial Decision Tree performed better than the Linear and Ridge baselines, with a lower RMSE and higher R² score.

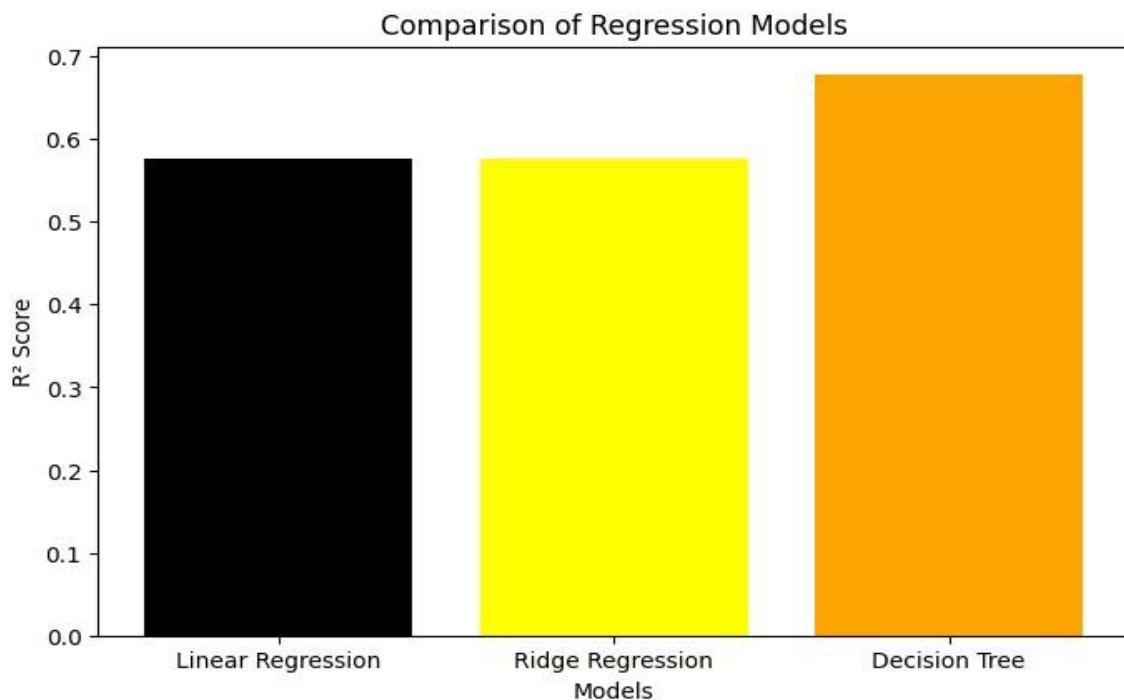


Figure 3. Initial R² comparison.

Validation, Hyperparameter Tuning and Final Conclusion Linear Regression

Validation:

5-fold CV scores: 0.6201, 0.6130, 0.6134, 0.6107, 0.6002

Average CV R²: 0.611484

Training R²: 0.612551 **Testing R²:** 0.575788

MAE: 0.533200 **RMSE:** 0.745581

The notebook reports that the Linear Regression model is generalizing well because the training-to-testing R² difference does not exceed the 0.10 overfitting threshold.

Decision Tree Validation:

MAE: 0.454679 **RMSE:** 0.703729 **R²:** 0.622076

Average 5-fold CV R²: 0.607227

Training R²: 1.000000 **Testing R²:** 0.622076

The unrestricted Decision Tree is flagged as overfitting because its training R² is 1.0 while the test R² is much lower.

GridSearchCV Tuning:

The notebook searches over **max_depth = [3, 5, 7, 10, None]** and **min_samples_split = [2, 5, 10]** using 5-fold cross-validation. The best parameters found are **max_depth = 10** and **min_samples_split = 10**, with a best CV R^2 of **0.695994**.

Tuned Decision Tree Results:

Test RMSE: 0.645688 Test R^2 : 0.681845

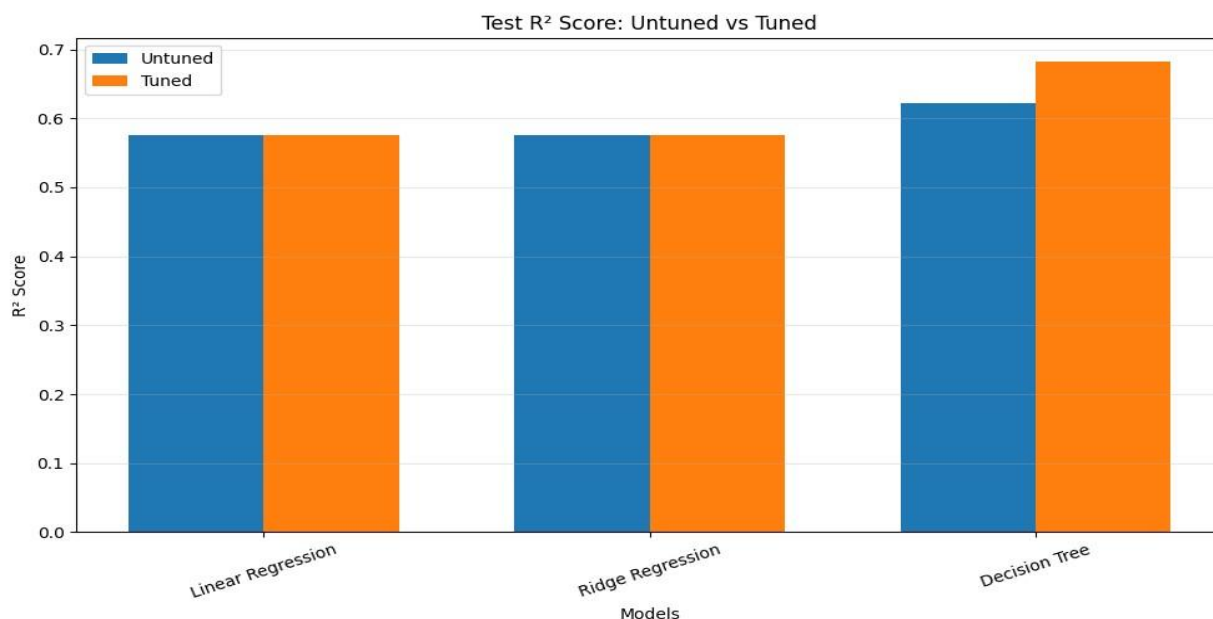


Figure 4. Untuned vs tuned model performance.

Final comparison

Model	R^2 Score	RMSE
Linear Regression	0.575788	0.745581
Ridge Regression	0.575855	0.745522
Decision Tree	0.622076	0.703729
Tuned Decision Tree	0.681845	0.645688

Conclusion: The tuned Decision Tree achieved the highest R^2 score and the lowest RMSE among the compared models, so it was selected as the final model. The trained model was saved as **house_price_model.pkl**.